

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Derive the expression for lift coefficient and explain its variation with angle of attack.
- Q.24 Discuss different types of drag in detail with diagrams.
- Q.25 Explain factors affecting longitudinal stability of an aircraft.

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2nd Sem/ Aeronautical Engg. Subject : Basics of Aerodynamics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The stratosphere is characterized by:
- a) Constant temperature
 - b) Increasing temperature with altitude
 - c) Decreasing temperature with altitude
 - d) High turbulence
- Q.2 Which dimensionless number compares inertial forces to gravitational forces?
- a) Reynolds number b) Froude number
 - c) Mach number d) Weber number
- Q.3 Circulation in fluid mechanics is related to:
- a) Viscosity b) Vorticity
 - c) Density d) Pressure

Q.4 Wingtip vortices primarily cause:

- a) Lift increase b) Induced drag
- c) Pressure drag d) Shock waves

Q.5 Sweepback of wings improves:

- a) Longitudinal stability
- b) Lateral stability
- c) Directional stability
- d) Lift coefficient

Q.6 Dynamic stability refers to:

- a) Initial tendency to return to equilibrium
- b) Oscillatory motion after disturbance
- c) Resistance to rolling
- d) Weathercock effect

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define troposphere.

Q.8 What is surface tension?

Q.9 State Lagrangian description of fluid motion.

Q.10 Define aerodynamic center.

Q.11 What is wave drag?

Q.12 Define dihedral effect.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the concept of atmospheric stability.

Q.14 Differentiate between laminar and turbulent flow.

Q.15 Define Reynolds number and explain its importance.

Q.16 Explain stagnation pressure and stagnation point.

Q.17 Describe symmetrical and cambered airfoils.

Q.18 Explain the concept of aspect ratio.

Q.19 Discuss subsonic transonic, supersonic and hypersonic regimes.

Q.20 Explain interference drag with examples.

Q.21 Describe static longitudinal stability.

Q.22 Explain keel effect in lateral stability.